UNIVERSITY OF KENT

KENT BUSINESS SCHOOL

2025--2026

Coursework Title: Data Analysis Report

Module: Machine Learning and Forecasting (BUSB7028)

Total marks:	100%
Weighting:	80%
Submission methods:	a) This assignment has two parts: Part A and Part B. They should be answered in one report and submitted to Individual Data Analysis Report. b) Only submit your PDF-formatted report. c) Do not print your Word-formatted file to a PDF file. Instead, you should save your Word-formatted file to a PDF file. The system does not accept a printed PDF file. d) The submission deadline is Wednesday, 18 March 2026, 12:00 noon.
Length:	a) The length of your assignment, including Part A and Part B, should be up to 2,500 words. b) Please do not copy and paste any part of this document onto your report.
Plagiarism:	See the module guide for a full description of the University's and the Business School's policies on plagiarism. The electronic version will be analysed using the *Turnitin* plagiarism-detection software. You will have the opportunity to submit drafts of your report to *Turnitin* where you will be able to see the originality report (but you may have to wait until the next day to see this). There will thus be no excuse for including any unreferenced material from other sources. You will probably include tables and graphs; it is **essential** that you reference the source at the bottom of every such entry. Plagiarism is most common when students copy and paste descriptive material from some source (and remember that doing this and then changing the occasional word still constitutes plagiarism). For more information, see https://student.kent.ac.uk/studies/plagiarism-and-how-to-avoid-it. The Business School expects students to use the Harvard system of referencing; you must read the university's guide https://www-citethemrightonline-com.chain.kent.ac.uk/category-list?docid=CTRHarvard. Please note that we are interested in what *you* can write about the information you have found. Please ensure the similarity score of your report is no more than 25%.

1. Part A

Answer Questions 1, 2 and 3.

Question 1.

Download dataset *sku_data.csv*. The dataset is sampled from a dataset in the UCI data repository, which implies that the dataset is a subset of the original dataset. You may find the description of the original dataset on https://archive.ics.uci.edu/dataset/585/stock+keeping+units (accessed on 06/Feb/2026). Cluster the dataset and explain your clustering process.

[10 marks]

Question 2.

A manufacturer, CX, started its business about many years ago. Recently, one of its products, *compact crane*, has become the leading product of CX, but the sales team has suffered with repeated over productions and under productions, due to wrong sales forecast. The production team needs to schedule how many compact excavators need to be produced one month in advance. Download dataset *TimeSeries.xlsx* for the sales data. Develop a time series forecasting model for CX and provide your modelling process.

[20 marks]

Question 3.

Download dataset *Regression_Dataset.csv*. Develop a linear regression model with variable *credit_score* as the dependent variable and the other variables as independent variables (you may select some of them), where "job_type1=1" represents "freelance", "job_type2=1" represents "fixed". Provide your modelling process and final model.

[20 marks]

2. Part B

[50 marks]

2.1 Problem and Requirements

❖ The problem

- You are expected to write a report, which effectively analyses and presents the following problem.

As a data analyst at our organization, you have been assigned to develop a classification model that predicts company bankruptcy. Bankruptcy classification should align with the business regulations of the relevant stock exchange

If we can develop a well-performing classification model, it will enable our organization to make more informed business decisions.

You will work with a historical dataset provided for this project. Your manager, Ms. Emma Simms, has already developed a logistic regression model as a starting point. She expects you to build a classification model with strong predictive performance—specifically targeting high metrics like accuracy.

Please write a data analysis report that meets the following requirements.

- Emma is an expert in *Machine Learning and Forecasting*. As such, to convince her to use your research findings, you may want to review the most advanced technique/algorithms and apply them into your data analysis.
- Since Emma did not give her logistic regression model to you, you must develop a logistic regression model for performance comparison.
- Emma also wants to assess which factors are most important to look out for and how they should impact the bankruptcy. Test your thinking against the data and show your findings in the report.

Note: The historical dataset Bankruptcy.csv can be downloaded from the module page on moodle.kent.ac.uk for the sales data.

- Columns beginning with `Bankrupt?' in the dataset show the bankruptcy, which has two values: 1=bankrupt and 0=not bankrupt. No further information about the variables is available.

❖ Requirements

- Part B should include up to **2000** words (+/- 10%, excluding abstract, references and appendices). The line spacing is 1.5 and the font size is 11.
- The word count (2000 words +/- 10%) does not include the list of references or appendices. The other parts of your report are included. You are expected to aim for this length, and while some minor variation is acceptable, the emphasis should remain on the quality of analysis and reporting rather than word count. Write clearly and concisely, selecting tables and figures thoughtfully based on their relevance to the content. If your report has much more words than the suggested maximum, it signals an implication that your report is not concise. Similarly, if it is significantly less than the suggested minimum, the coverage of your report is insufficiently comprehensive.
- Use the Harvard referencing style.

2.2 Learning outcome, checklist, and note

Intended learning outcome – to develop and practice the critical skills required to conduct a data analysis project.

Checklist

When performing a data analysis project, you need to ask a series of technical and business questions about the content, presentation, and detail of your report. Use the following questions to help guide your data analysis process. They are just a start. You may have other issues you wish to include.

❖ Contents

- What will your project really want to accomplish?

- What are the criteria for a successful or useful outcome to your project?
- What are the other factors that should be considered in determining the data analysis goal?
- Why do you think that the organisation can benefit from your data analysis outcomes?
- Is there any other business that may benefit from your report?
- What is your judgement about the importance of this work?

❖ **Data analysis technique/process**

- How do you select data analysis techniques? Why?
- Have you missed out any key assumptions that you should check in your models?
- Have you researched any other modelling techniques that are more advanced and more efficient?
- Are the conclusions supported by the data analysis presented?
- Have you checked the quality and reliability of your data?
- Have you presented and interpreted your modelling results clearly and correctly?

❖ **Report structure**

- Do you have an introduction section? Is the introduction clear and well written?
- Do you have a conclusion section? Are your conclusions clear and logically correct?
- Is it well-organised, in that there is a logical flow of information that leads to a supportable conclusion(s)?
- Are the figures and tables well presented – do they help? Might the results have been better presented and if so, how?

❖ **Writing**

- Do you assume the reader of your report an expert or a layperson?
- Have you invited a friend of you to proofread the language used in the report? Are you sure your report is free of grammatical or typing errors?
- If your colourful figures will be printed in black and white, are you sure that the data presented in each figure are distinguishable?
- Have you listed your references out?
- Have you cited your references properly?
- Does each figure/table in your report have a numbering and a caption? Have you cited them with their numberings? Have you interpreted each figure/table?
- Does each section in your report have a numbering?

Note

- Any uninterpreted figures, tables, or equations will not be marked.
- Do not simply copy and paste Python interface/outputs onto your assignment without any interpretation. The focus of this assignment is not on how well you use a data analysis package.

2.3 Section C: Marking criteria

Marking Criteria

Explanations of how the generic criteria will be interpreted for this particular report are shown in **purple**

Mark	Reading & Knowledge	Understanding, analysis and argument	Organisation and presentation
100 95 85	Very extensive reading and exceptionally comprehensive knowledge which indicates a high level of independent background research • Enterprising research into modelling techniques • Enterprising research into data sources • Excellent appreciation of the relevance of the data to the context	Very full and perceptive awareness of subject matter, with original critical and analytical assessment of the issues and excellent grasp of their wider significance. Combines a comprehensive understanding of theoretical issues and empirical application. Clear evidence of a high level of originality and independent thought along with an ability to defend a position logically and convincingly with arguments presented that are sophisticated and highly challenging • Excellently applied most advanced/latest data analysis algorithms and built a model that may be trusted and used by your client • Excellently written commentary and interpretation which shows genuine original thought • Excellent appreciation of relationship between ideas • Very clear conclusions showing critical appraisal and depth of analysis	Very well structured piece of work demonstrated by an excellent arrangement & development of material and argument. Excellent English and meticulous presentation, with immaculate referencing and extensive bibliography • All parts present, linked and clearly labelled and with an excellently thought through argument • Outstanding introduction, perfect coverage which captures the reader's interest • Outstanding presentation and description of the data • Illustrations/graphs/tables: excellent design, eye-catching
78 75 72	Wide reading and comprehensive knowledge of the subject matter which goes beyond the course material and lectures • Very good research into modelling techniques	Full and perceptive awareness of issues, demonstrating an appreciation of all major points and a clear grasp of their wider significance. Clear evidence of independent thought, ability to defend a position logically and convincingly. At the higher end can make linkages between theoretical issues and empirical application and also demonstrates some originality and flair • Very well applied advanced data analysis algorithms. In the commentary and interpretation, develops original ideas and conveys them well and built a model that may be trusted and used by your client	Well-structured piece of work with very careful thought given to arrangement and development of material and argument. Excellent English with appropriate referencing and comprehensive bibliography. • All parts present, linked and clearly labelled • Excellent coverage in introduction, clear and well written • Very good presentation and description of the data

	• Very good research into data sources • Very good appreciation of the relevance of the data to the context.	• Clear appreciation of relationship between ideas • Clear conclusions that follow logically from earlier analysis	• Illustrations/graphs/tables: very effective, good design
68 65 62	Good range of reading and good knowledge which comprehensively covers course material and may go beyond this • Good research into modelling techniques • Good research into data sources • Good appreciation of the relevance of the data to the context.	A comprehensive argument demonstrating a sound awareness of issues and a serious understanding of their wider significance. Evidence of careful thought with a well developed argument. Use of theories, concepts and research findings which are largely precise although there may be minor factual errors or inaccuracies • Well applied relevant data analysis algorithms • In the commentary and interpretation, develops clear ideas • Shows how ideas can relate to each other • Conclusions follow from main body of report, but may miss some elements or introduce new topics	Well-organized answer with a clear structure which facilitates the development of material and argument. Good English with appropriate referencing and bibliography. • Headed sections and logical connection between them • Introduction covers everything that is to come in the report but not perfectly expressed • Everything relating to the data necessary for the reader's understanding has been presented but maybe lacking extra detail • Illustrations/graphs/tables: effective and correctly presented
58 55 52	Fair range of reading and reasonable knowledge of course material • Competent research into data sources • Narrower appreciation of the relevance of the data to the context.	Essay is a satisfactory response to the question. It demonstrates some grasp of theory and its relation to empirical data but less awareness of issues and their wider significance. It is largely accurate but may contain more than minor errors such as incorrect facts or errors of omission. There may be some attempt at serious argument but this is not fully developed. The essay may appear correct but lacks originality and clarity of though for an above average grade. • Adequately applied relevant data analysis algorithms • In the commentary and interpretation, covers the essentials, but does not develop ideas or the linkages between them	Adequate effort to organise the material and argument. Adequate English with reasonable referencing and bibliography • Must have headed sections but with limited flow of argument • Introduction says what is in the report but lacks clarity or sufficient detail • Presentation and description of the data clear but lacking inspiration or interesting detail

		• Some conclusions, but missing some key elements of what has been presented before	• Illustrations/graphs/tables: presents essential information at a basic level
48 45 42	Limited reading and incomplete knowledge of course material • Limited research into modelling techniques • Limited research into data sources, • Bare appreciation of the relevance of the data to the context. • Shows a lack of appropriate effort	Limited awareness of issues and of their wider significance. The information it contains is insufficient and theories and concepts used may contain errors, omissions and irrelevancies. Limited thought and argument. • Limited application of relevant data analysis algorithms • Basic and limited commentary, not going beyond minimum • No real appreciation of relationships between ideas • Very basic conclusions	Structure of the essay is weak and there is a limited effort to organise material and argument. Satisfactory English but with limited referencing and bibliography • Some structure, discernible separation of parts but no useful headings • Introduction covers what is to come, but only in barest form • Adequate amount of data presented, but appreciable gaps and/or poorly written description • Illustrations/graphs/tables: shows some information but poorly presented and difficult to understand
38 35 32	Very limited reading and knowledge with evident gaps in knowledge. Lack of familiarity with course material is evident • Very limited research into modelling techniques • Very limited research into data sources • Virtually no appreciation of the relevance of the data to the context. • A serious lack of effort.	Very limited awareness of issues and of their wider significance and contains a small amount of acceptable and relevant information. Clearly deficient answer which contains substantial errors, omissions or irrelevancies • Inadequate application of relevant data analysis algorithms • Some commentary and interpretation but much missing or presents difficulties in understanding • Almost nothing within conclusions linking to earlier analysis	Structure is largely absent indicating very little effort put into organising material. Significant errors in English and poor referencing and bibliography • Poorly structured, no sections, little connection between parts • Introduction contains very few words or is an inadequate description • Can just see something of potential interest in the data but no proper description • Illustrations/graphs/tables: minimal, very poorly presented
25 20 10	Very poor level of reading and failure to demonstrate competent knowledge.	Lacks familiarity with the subject, demonstrating very poor awareness of issues and of their wider significance.	Essay is structureless and there is a clear lack of organisation of material. Substantial errors in English and very poor or absent referencing and bibliography.

0	Knowledge of course material is completely absent • No real effort shown to research into modelling techniques • No real effort shown to identify relevant data and appreciate the context	Competent understanding is absent and essay is characterised by confusion • No application of relevant data analysis algorithms • Very little or no intelligible commentary or interpretation • No proper conclusions	• No discernible structure • No or very minimal introduction • Little or nothing of any substance in description of the data • Illustrations/graphs/tables: none, illegible, unintelligible